

A Novel Sensor System for Liquid Properties Based on a Micromachined Beam and a Low-Cost Optical Readout

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Abstract—For the measurement of liquid parameters like viscosity and density vibrating micromachined structures offer useful alternatives to bulky conventional measurement equipment. Evaluating the shift of the resonance frequency and the change of the damping factor of such devices allows the simultaneous determination of viscosity and density. Furthermore, these sensors can even be used for complex liquids like emulsions where other microacoustic sensors like TSM quartz resonators, in comparison to conventional laboratory viscometers, measure in a different rheological domain. In our contribution we present a novel implementation of a viscosity and density sensor system utilizing a doubly-clamped vibrating micromachined beam. The beam deflection is determined by means of a laser pickup head as it is used, e.g., in DVD drives, yielding high sensitivity and a low-cost setup at the same time. The measured damping factor correlates well with the viscosity of the respective liquid, whereas the resonance frequency is also influenced by the liquid's density.

I. INTRODUCTION

Micromachined cantilevers have been successfully used as liquid property sensors [1]. The evaluation of the resonance frequency and the quality factor of a vibrating cantilever allows the simultaneous determination of viscosity and density [2]. In general, such sensors also feature lower resonance frequencies than, e.g., quartz thickness shear-mode (TSM) resonators and are therefore more suitable for sensing the viscosity of non-Newtonian and complex liquids [3].

We have recently presented a micromachined doubly clamped vibrating beam device [4] (Fig. 1) which can be utilized as a liquid property sensor. The beam vibrations are excited by Lorentz forces. When the sensor is immersed in a liquid, both the resonance frequency f_r and the damping factor D (the reciprocal of the quality factor) of the device are changed depending on the liquid parameters, i.e., density and viscosity. Therefore, a suitable measurement method aims at determining f_r and D , and thus requires the measurement of the actual beam deflection. In principle, the change of the effective device impedance due to piezoresistive and induction effects can be used to measure the actual deflection of the

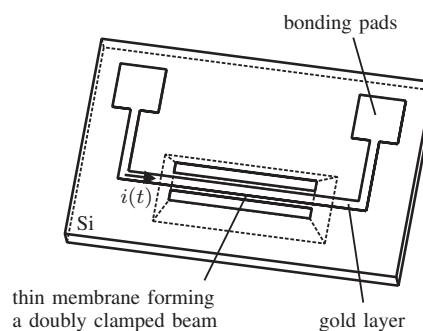


Fig. 1. Doubly clamped beam device featuring a micromachined silicon nitride (Si_3N_4 , SiN_x) beam and a conductive path (gold layer) allowing for Lorentz force excitation. The beam has a length of $350\text{ }\mu\text{m}$, a width of $40\text{ }\mu\text{m}$, and a thickness of $1.3\text{ }\mu\text{m}$.

beam [4]. However, for the considered micromachined design these effects do not provide the necessary sensitivity for measuring the small deflection amplitudes expected for immersion in a liquid. Therefore, a readout featuring higher sensitivity is required.

In this contribution, we present a novel sensor system based on a micromachined clamped-clamped vibrating beam and a low-cost optical readout utilizing a DVD pickup head. The measurement setup is described in detail. Finally, measurement results demonstrate the feasibility of the sensor system.

II. SENSOR FABRICATION

The doubly clamped beam sensor was fabricated by coating a $350\text{ }\mu\text{m}$ thick (100) silicon wafer with 320 nm silicon nitride (Si_3N_4) on both sides. Next, a Ti-Au-Cr layer has been deposited and structured to form a conductive path over the doubly clamped beam and the bonding pads. A low stress silicon nitride (SiN_x) protective film of 1000 nm has been applied using a low temperature PECVD process. After creating the apertures for the bonding pads and the backside etch window by means of reactive ion etching, a thin membrane has been

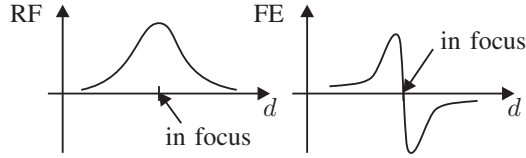


Fig. 2. RF and FE (focus error) signals of a CD/DVD pickup unit vs. distance d between the pickup and the device under test.

manufactured by means of KOH based anisotropic backside wet etching. In order to obtain a clamped-clamped beam the membrane has subsequently been structured from the front side by reactive ion etching. Finally, the chromium has been removed from the bonding pads by means of wet-etching. The doubly clamped beam features a thickness of $1.3\ \mu\text{m}$, a length of $320\ \mu\text{m}$, and a width of $40\ \mu\text{m}$ (Fig 1).

III. OPTICAL READOUT

To obtain the resonance frequency and the damping of a micromachined structure the measurement of the actual beam deflection is required. This could be achieved by providing, e.g., piezoelectric or piezoresistive layers on the beam, or designing structures for capacitive readout. For the beam excited by Lorentz forces also an inductive readout could be utilized due to the presence of a permanent magnetic field. The beam represents a conductive layer moving in a magnetic field $\vec{B}$. The voltage thereby induced leads to a change of the effective impedance of the beam [4]. However, for the considered design the change of the effective impedance is in the range of $30\ \text{m}\Omega$ for the beam vibrating in air. When immersed in liquids, the vibration amplitudes and thus the change of impedance decrease. Consequently, the measurement results suffer from a low signal-to-noise ratio.

Optical readout methods have been successfully used for micromachined sensors in liquids. A reflective sensor and evaluation of the signals by a synchronous demodulator [3] offer a cheap means for the measurement of micromachined beam deflection. However, for the small deflection amplitude of the presented doubly clamped beam, a more sensitive readout is required. The optical pickup heads of DVD drives offer a highly integrated and low-cost alternative [5]. For the purpose of focus control the pickup offers a focus error (FE) output signal, that depends on the distance between the pickup lens and an opposite surface (Fig. 2). Furthermore, the RF output signal of the pickup, which is normally used to read the data stream from a DVD, is also distance dependent. Together with a customized electronics both signals can be used to measure out-of-plane deflections of a vibrating beam. The optical readout features high sensitivity and high bandwidth. As a trade-off optical readout limits the use of the sensor to liquids that are transparent at the pickup laser's wave length.

IV. MEASUREMENT SETUP

Figure 3 shows a photo of parts of the prototype setup. The vibrating doubly clamped beam sensor is glued on an carrier which also provides the electric contacts, and is immersed in

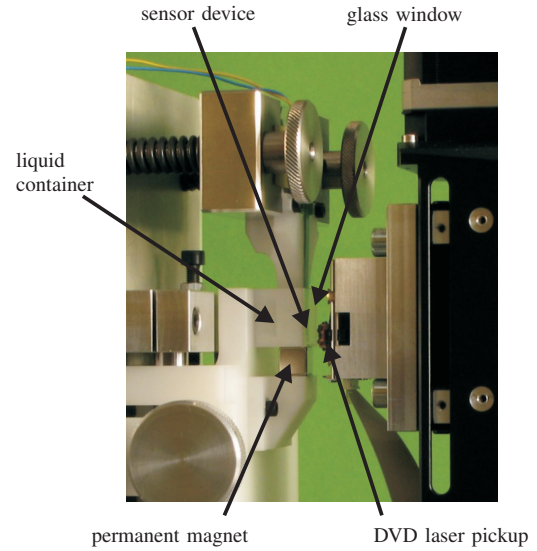


Fig. 3. Prototype setup with the micromachined device immersed in the sample liquid. The static magnetic field is excited by the permanent magnet below.

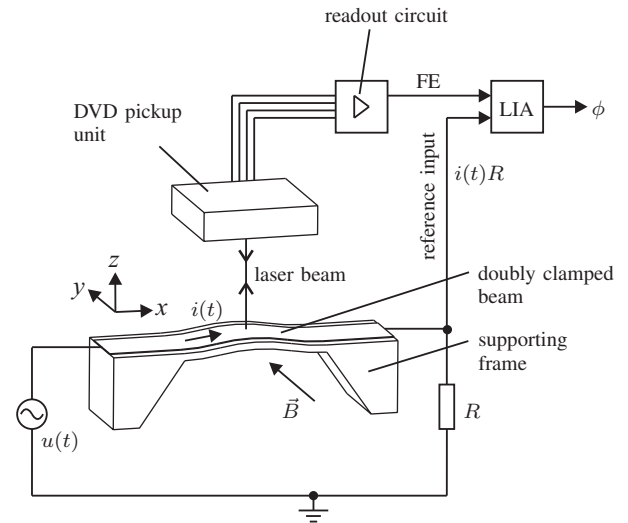


Fig. 4. Schematic diagram of the sensor system. The beam vibrations are excited by Lorentz force caused by $i(t)$ and the magnetic field $\vec{B}$. From the FE signal the actual beam deflection in z -direction is obtained. The phase shift ϕ between the excitation current and FE (beam deflection) is measured by a lock-in amplifier (LIA).

a sample liquid container. At the bottom side of the liquid container a permanent magnet has been attached to provide the required magnetic field ($B_y \approx 100\ \text{mT}$). On the front side the liquid container has a transparent window, allowing for optical readout.

A schematic of the setup is given in Fig. 4. Beam vibrations are excited by applying a constant magnetic field (in y -direction) and a sinusoidal current (x -direction) to the device. The DVD pickup's laser beam is focused on the center of the vibrating structure. For positioning of the laser beam during the experiments, the pickup is mounted on a 3D translation

TABLE I
REFERENCE VALUES FOR VISCOSITY η AND DENSITY ρ OF THE SAMPLE LIQUIDS

sample liquid	η [mPa · s]	ρ [kgm ⁻³]
Toluene	0.565	867
Cyclohexane	0.895	779
Ethanol	1.056	789
1-Propanol	1.898	803
1-Butanol	2.524	810
1-Pentanol	3.441	814
1-Hexanol	4.486	814
1-Heptanol	5.942	822
1-Octanol	7.368	827

The reference values for the viscosity and density have been obtained from [6] (25°C temperature) and [7], respectively.

stage featuring a position repeatability of 100 nm. However, in a commercial application the optical pickup head could be mounted at a fixed position, thus not requiring such sophisticated equipment. A dedicated readout electronics controls the laser power and obtains the distance dependent FE signal. Subsequently, a lock-in amplifier (LIA) determines the phase shift ϕ between the beam current $i(t)$ and the beam deflection in z -direction given by FE. From a frequency sweep $\phi(j\omega)$ both the resonance frequency f_r and the damping D can be obtained.

V. RESULTS

To demonstrate the feasibility of the sensor system, a number of measurements have been carried out. As sample liquids, a variety of alcohols and other pure chemicals have been used: ethanol, 1-propanol, 1-butanol, 1-pentanol, 1-hexanol, 1-heptanol, 1-octanol, cyclohexane, and toluene. They exhibit viscosities in the range of 0.565 to 7.368 mPa·s at 25°C (Tab. I).

The doubly clamped beam sensor was immersed in the respective sample liquid, and an excitation current of 3.1 mArms was applied. A frequency sweep measurement yields the device's frequency characteristics $\phi(j\omega)$. Figure 5 shows the phase shift frequency characteristics of the beam near the first mode of resonance (dotted lines). To obtain the resonance frequency $f_r = \omega_r/(2\pi)$ and the damping factor D , a second order transfer function

$$\phi(j\omega) = \arg \left\{ \frac{1}{1 + j\omega \frac{2D}{\omega_r} - \frac{\omega^2}{\omega_r^2}} \right\} \quad (1)$$

was fitted to the measurement results (solid lines in Fig. 5).

To first order, the damping factor D of a vibrating cantilever is dominantly influenced by the viscosity of the respective liquid [8], which is confirmed by the results given in Fig. 6. Figure 7 shows the resonance frequency shift $(f_r - f_{r,air})/f_{r,air}$, where $f_{r,air} = 96.7$ kHz is the beam's resonance frequency in air. The results indicate that the resonance frequency of the vibrating beam is not only influenced by the liquid's viscosity but also by the density. Compared to the results obtained for cyclohexane, the measurement of a beam immersed in

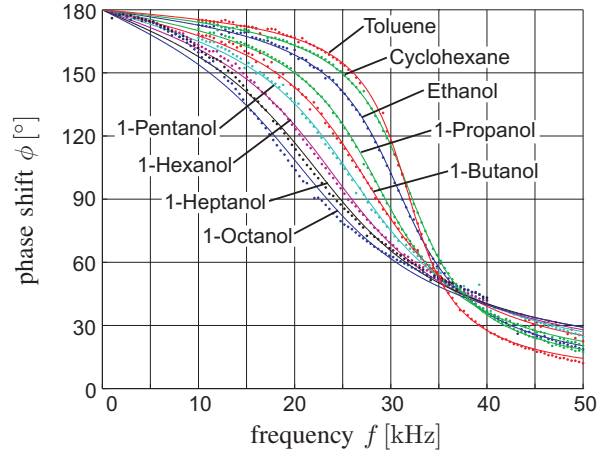


Fig. 5. Phase shifts near resonance between the excitation current $i(t)$ and the actual deflection of the beam immersed in various sample liquids (dotted lines). The phase shift provides a better signal to noise ratio than the amplitude characteristics, and is thus used to extract the resonance frequency and the damping by curve fitting (solid lines).

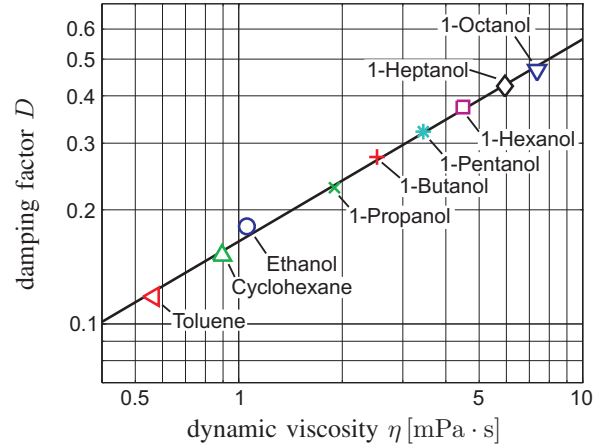


Fig. 6. Measured damping D of the vibrating beam vs. dynamic viscosity η of the sample liquids.

toluene results in an additional frequency shift due to the higher density of the liquid (Tab. I).

VI. CONCLUSION

The resonant behavior of a doubly clamped micromachined vibrating beam is influenced by the surrounding liquid. In particular, the measurement of the resonance frequency and the damping factor allows the determination of the viscosity and the density of the liquid. For this purpose, the actual beam deflection must be obtained. The presented prototype setup utilizes a low-cost optical readout based on a DVD laser pickup head and a customized electronics. The measurement results show that the obtained damping factor is mainly dominated by the liquid's viscosity, whereas the resonance frequency is also influenced by the density of the respective liquid. Therefore the sensor system can directly be used to determine the viscosity of a liquid. This has been successfully demonstrated for liquids exhibiting viscosities in the range of

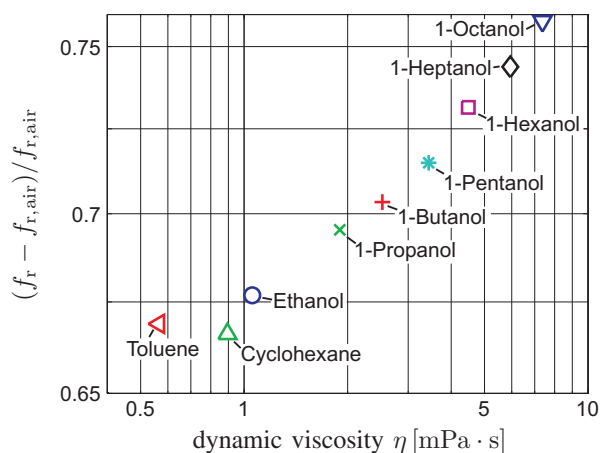


Fig. 7. Shift of the resonance frequency of the vibrating beam vs. dynamic viscosity η of the sample liquids. The results indicate that the resonance frequency is not only influenced by the viscosity of the respective liquid, but also by its density.

0.565 to 7.368 mPa·s. However, to measure the liquid's density simultaneously, a suitable model for the sensor device must be applied, e.g., [9].

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REFERENCES

- [1] P. I. Oden, G. Y. Chen, R. A. Steele, R. J. Warmack, and T. Thundat, "Viscous drag measurements utilizing microfabricated cantilevers," *Appl. Phys. Lett.*, vol. 68, pp. 3814–3816, 1996.
- [2] L. F. Matsiev, "Application of flexural mechanical resonators to simultaneous measurement of liquid density and viscosity," in *1999 IEEE Ultrasonics Symposium*, 1999, pp. 457–460.
- [3] A. Agoston, F. Keplinger, and B. Jakoby, "Evaluation of a vibrating micromachined cantilever for measuring the viscosity of complex organic liquids," *Sens. Actuators A*, vol. 123–124, pp. 82–86, 2005.
- [4] E. K. Reichel, B. Weiß, C. Riesch, A. Jachimowicz, and B. Jakoby, "A novel micromachined liquid property sensor utilizing a doubly clamped vibrating beam," in *Proc. Eurosensors XX*, Göteborg, Sweden, Sept. 17–20, 2006.
- [5] N. Scuor, P. Gallina, O. Sbaizero, H. V. Panchawagh, and R. L. Mahajan, "Dynamic characterization of MEMS cantilevers in liquid environment using a low-cost optical system," *Meas. Sci. Technol.*, vol. 17, pp. 173–180, 2006.
- [6] D. S. Viswanath, T. K. Ghosh, D. H. L. Prasad, N. V. K. Dutt, and K. Y. Rani, *Viscosity of Liquids*. Springer, 2007.
- [7] D. R. Lide, Ed., *CRC Handbook of Chemistry and Physics*, 83rd ed. CRC Press, 2002.
- [8] C. Riesch, E. K. Reichel, F. Keplinger, and B. Jakoby, "Characterizing resonating cantilevers for liquid property sensing," in *Proc. IEEE Sensors*, Daegu, Korea, Oct. 22–25, 2006, pp. 1070–1073.
- [9] C. A. V. Eysden and J. E. Sader, "Small amplitude oscillations of a flexible thin blade in a viscous fluid: Exact analytical solution," *Physics of Fluids*, vol. 18, p. 123102, 2006.